Exam Ace Champs

Where Champions Are Made

Quick Revision

Chapterwise

Mind-Maps

BIOLOGY

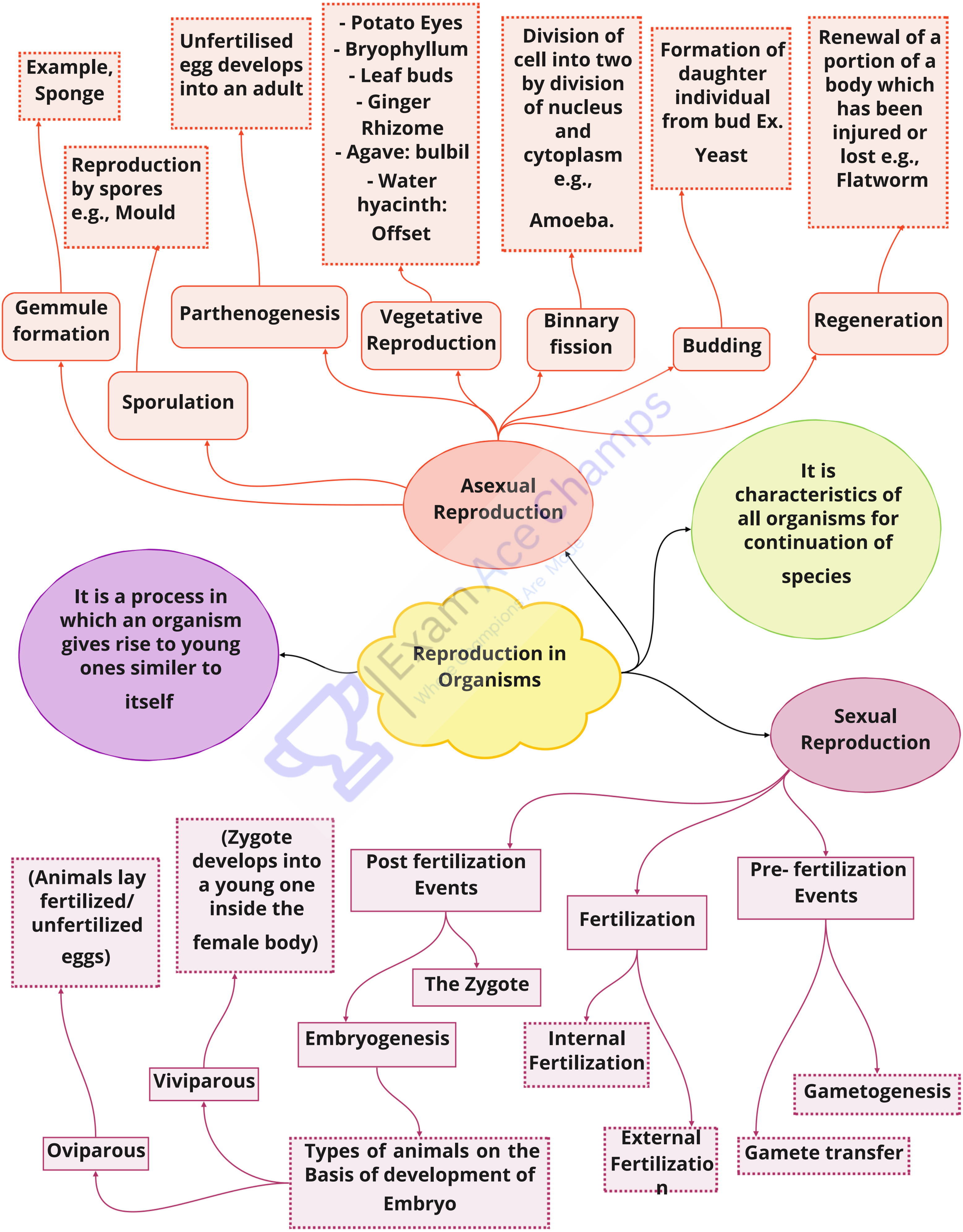
class 12

Covers all chapters of NCERT

Faster Recall

Quick Revision

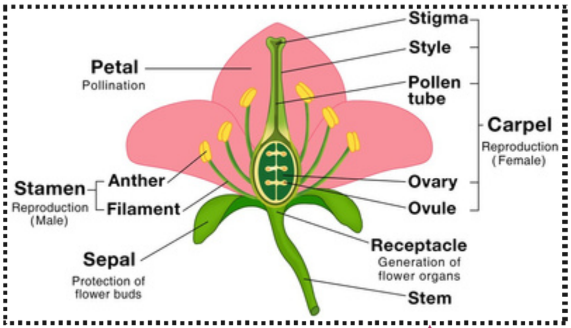
Class : 12th Biology
Chapter- 1 : Reproduction in Organisms



Class : 12th Biology
Chapter- 2 : Sexual Reproduction in Flowering Plants

Sexual Reproduction in Flowering Plants

Flower Structure

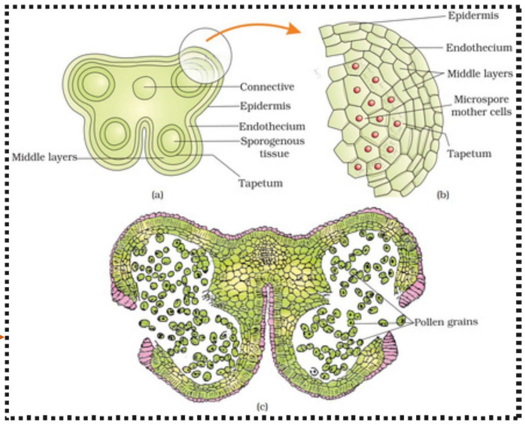


Significance of seed dispersal and fruit formation

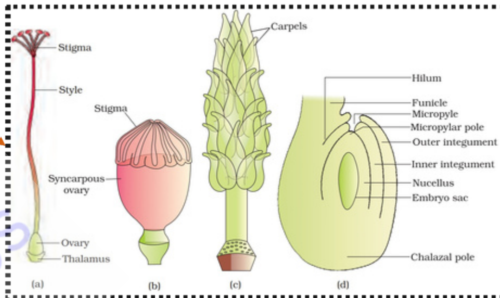
Seed dispersal help the species to colonize in other areas and many fruits have mechanisms for dispersal of seeds

Pre- fertilisation events

Male gametophyte

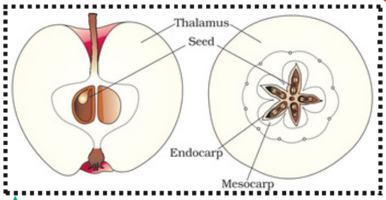


Female gametophyte



Special modes

- Polyembryony – Occurrence of more than one embryo in a seed
- Apomixis – Production of seed without fertilization
- Parthenocarpy – Development of fruit without fertilization

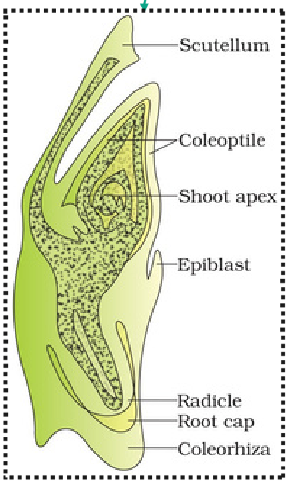


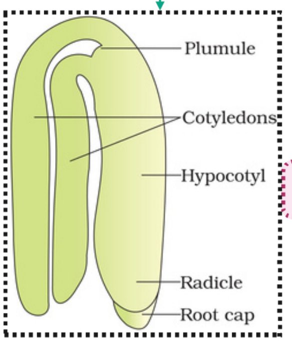
Pollination

- Types
  - Autogamy
  - Xenogamy
  - Geitonogamy
- Agencies
  - Abiotic
    - Wind
    - Water
  - Biotic
    - Animals
- Example
  - Winter Pollinated: Grasses Water Pollinated: Vallisneria and Hydrilla Insect Pollinated: Yucca

Post- fertilisation Events

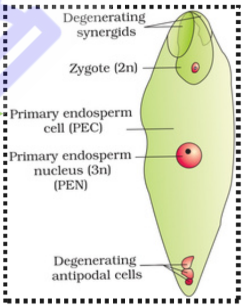
- Formation of fruit
  - False Fruit – When parts other than ovary contribute to the fruit formation. Ex : Apple
  - True Fruit – When fruit develops from ovary Ex : Mango
- Development of endosperm
- Development of embryo
  - L.S. of Monocot embryo
  - Dicot embryo
- Development of Seed
  - Albuminous seeds (Endospermic)
  - Non- albuminous seeds (Non- endospermic)





Double fertilization

Syngamy + Triple fusion = Double fertilization.



Pollen- pistil interaction

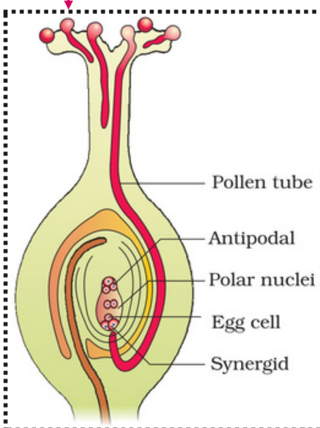
It is a dynamic process involving pollen recognition followed by pollen tube growth

Artificial Hybridisation

It is one of the major approaches of crop improvement by using desired pollen grains for pollination.

Achieved by

- Emasculation
- Bagging



Outbreeding Devices

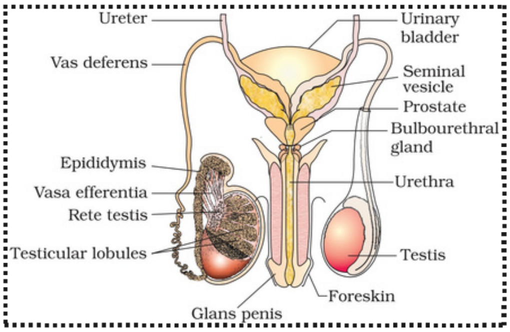
Devices for promoting cross pollination

1. Self–incompatibility
2. Arrangement of anther and stigma at different positions.
3. Production of unisexual flowers (Decliny)
4. Pollen release and stigma receptivity not synchronised.

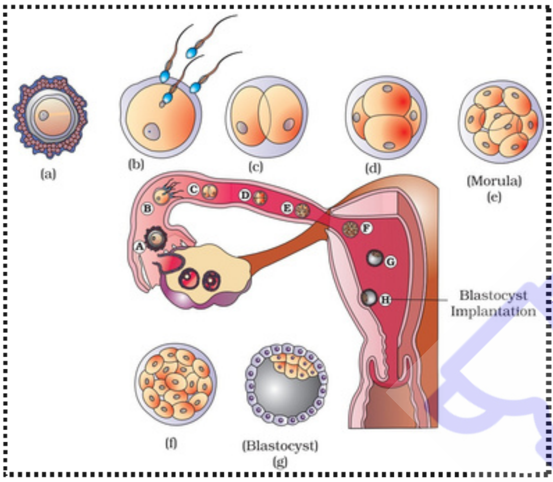
Class : 12th Biology
Chapter- 3 : Human Reproduction

Human Reproduction

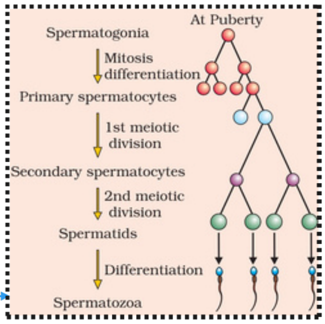
- **Male Reproductive System**
  - External Genitalia
  - External Ducts
    - Rete testis, vasa efferentia, Epididymis & Vas deferens
  - External glands
    - Seminal vesicle, prostate, Bulbourethral gland
  - Testes
    - is made of
      - Seminiferous tubules
        - Consist of
          - Spermatogonia, Sertoli cells and Leydig cells

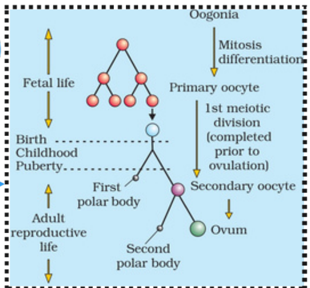


- **Parturition**
  - Process of giving birth to young ones after gestation period
- **Female Reproductive System**
  - Mammary Glands
    - Mammary lobes
    - Mammary duct
    - Lactiferous duct
  - External Genitalia
    - Mons pubis
    - Labia majora
    - Labia minor
    - Hymen
    - Clitoris
  - Accessory Ducts
    - Oviduct
      - Infundibulum
      - Ampulla
      - Isthmus
    - Vagina
    - Uterus
  - Ovarian Stroma
    - Ovaries
      - Ovarian Stroma
        - Cortex
        - Medulla
- **Lactation**
  - Production of milk towards the end of pregnancy
- **Blastocyst implantation**

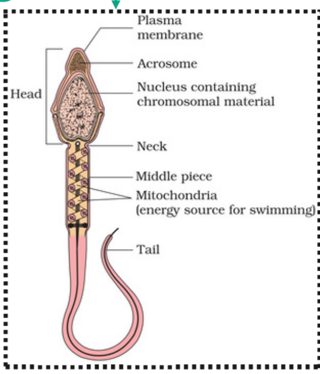


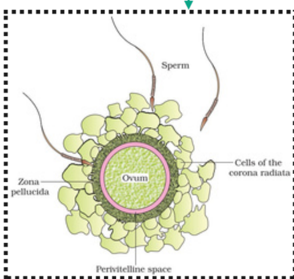
- **Gametogenesis**
  - Spermatogenesis
  - Oogenesis



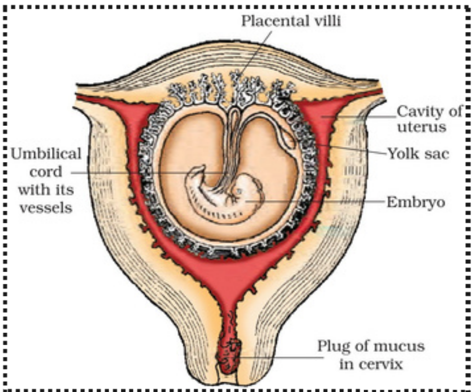


- **Fertilization**
  - Fusion Between sperm and ova to form zygote

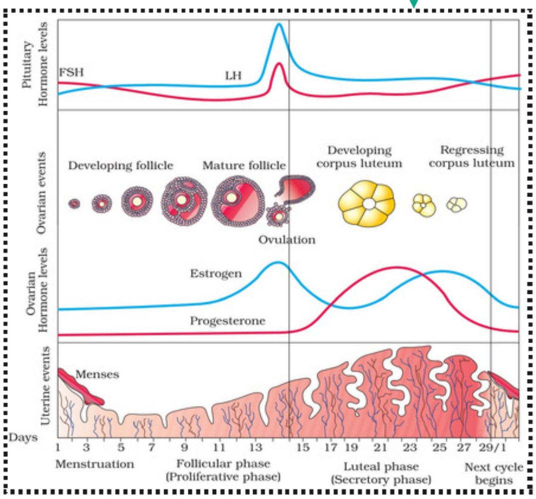




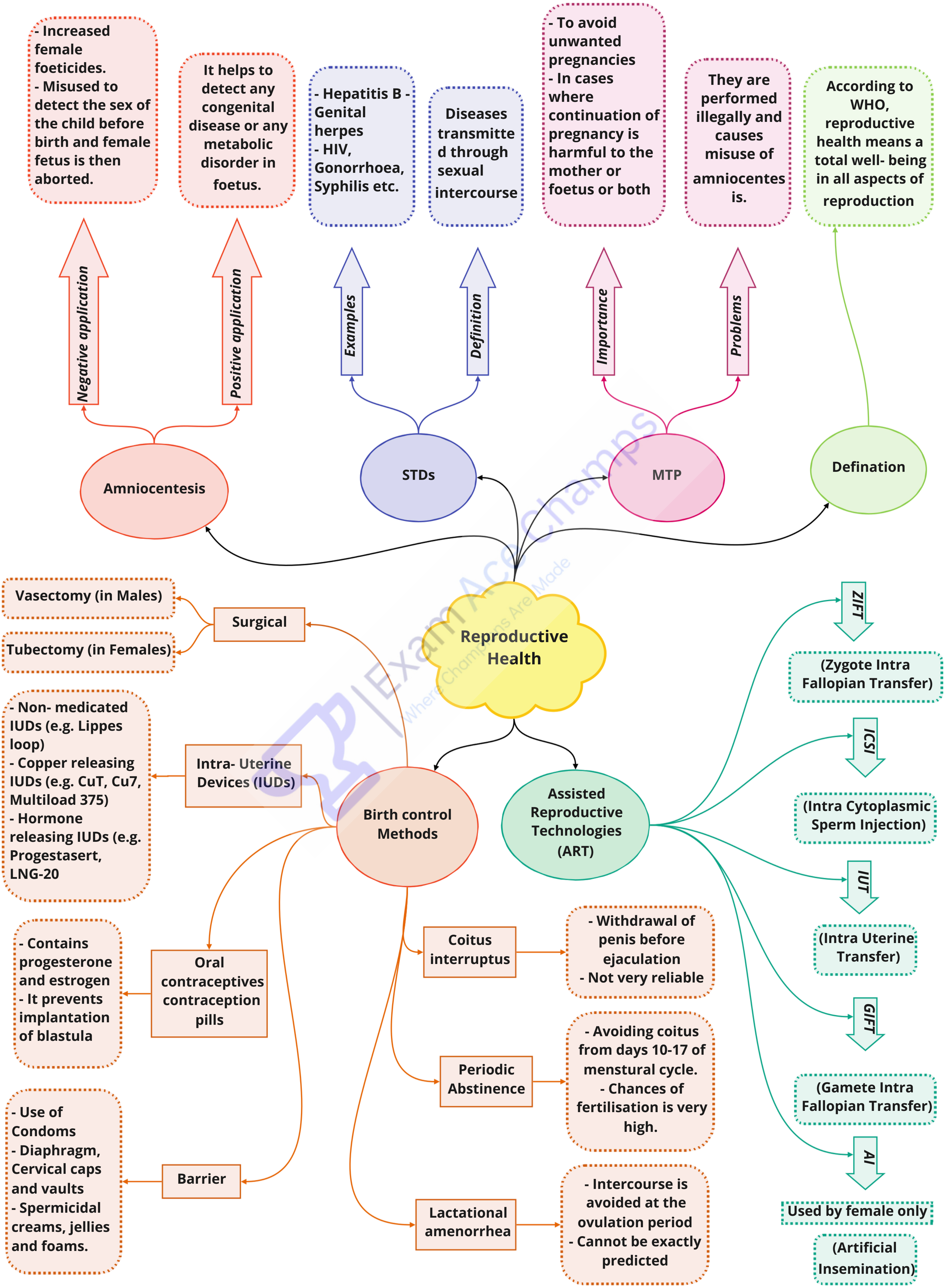
- **Pregnancy and placenta formation**
  - hPL
  - hCG
  - Estrogen
  - Progestogen



- **Mestrual cycle**



Class : 12th Biology
Chapter- 4 : Reproductive Health



Class : 12th Biology
Chapter- 5 : Principles of Inheritance and Variation

Principles of Inheritance and Variation

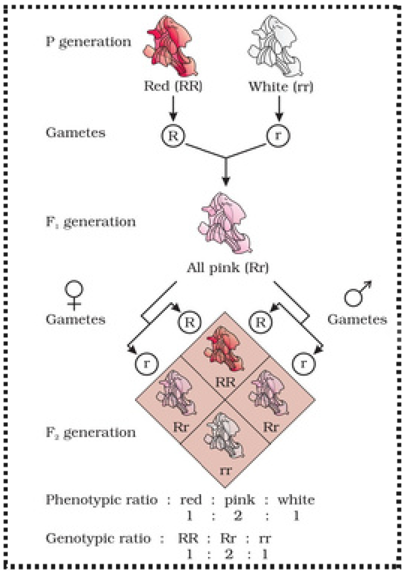
Gregor Mendel

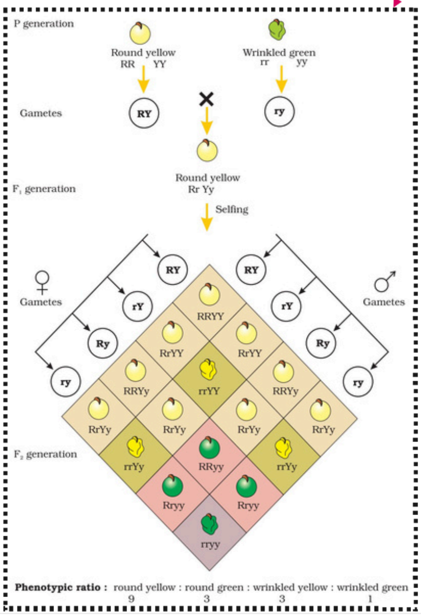
Studied 7 Pairs of contrasting characters of pea plant

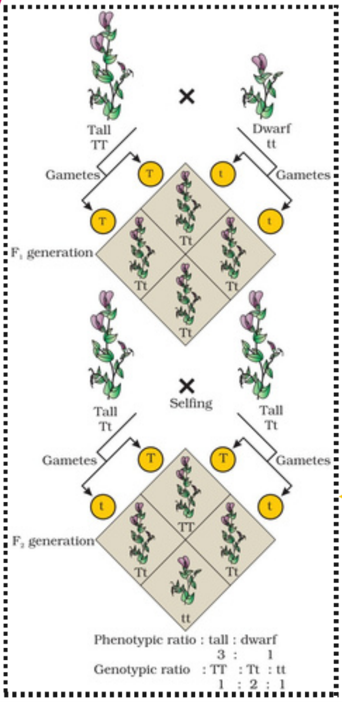
S.No.	Characters	Contrasting Traits
1.	Stem height	Tall/dwarf
2.	Flower colour	Violet/white
3.	Flower position	Axial/terminal
4.	Pod shape	Inflated/constricted
5.	Pod colour	Green/yellow
6.	Seed shape	Round/wrinkled
7.	Seed colour	Yellow/green

Proposed three laws

- **Law of Dominance**: In heterozygous condition only one member of a pair expresses its effect in the hybrid and is called as dominant while the manifestation of the other is masked and is known as recessive
- **Law of Segregation**: During gamete formation, the factors (alleles) of a character pair present in parents segregate from each other such that a gamete receives only one of the two factors
- **Law of Independent assortment**: When two pairs of traits are combined in a hybrid, segregation of one pair of characteristics is independent of the other pair of characters.





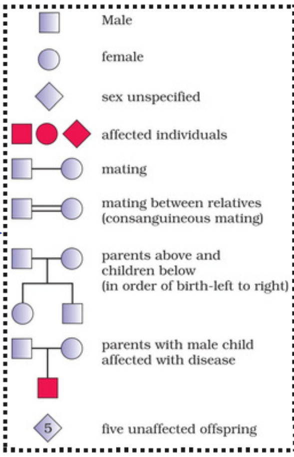


Genetic Disorder

- **Mendelian Disorders**
  - Autosomal dominant: E.g. Muscular dystrophy.
  - Autosomal recessive: E.g. Sickle cell anaemia, Albinism
  - Sex linked: E.g. Haemophilia
- **Chromosomal Disorders**
  - Example, Down's syndrom (trisomy of 21).
  - Klinefelter's Syndrome (XXY in male).
  - Turner's syndrome (XO in female).

Pedigree Anlysis

- **Definition**: Record of occurrence ofatrait in several generations of a family
- **Symbol used**



- **Importance**: Pedigree study is utilized to trace the inheritance of a specific trait, abnormality or disease

Mutation

- **Definition**:
  - Sudden heritable change in DNA sequence, which results in changes in the genotype and the phenotype of an organism.
  - Leads to variation in DNA
- Frameshift Mutation
- Point Mutation

Sex Determination

- ZZ— ZW mechanism Example, Birds
- XX— XO mechanism Example, grasshopper
- XX – XY mechanism Example, Human being

Chromosomal Theory of inheritance

- Proposed by Walter Sutton and Theodore Boveri in 1902.
- Thomas Hunt Morgan formulated chromosomal theory of inheritance using fruit flies (Drosophila melanogaster).
- Morgan coined the term Linkage.

Non- Mendeliam Inheritance

- **Incomplete Dominance**
- **Co- Dominance**: Two alleles of a gene are equally dominant and express themselves even when they are together. - E.g. ABO blood grouping in human
- **Pleiotrophy**: Ability of a gene to have multiple phenotypic effects as it influences a number of characters simultaneously
- **Multiple Allels**: A gene exists in more than two allelic forms E.g. AB , O blood grouping
- **Incomplete Dominance**: One of the two alleles of a gene is incompletely sp dominant over the other allele. E.g., Flower colour in Antirrhinum sp. and Mirabilis jalapa.

Molecular Basis of Inheritance

Polynucleotide Chain

- Made of Nucleotide
  - Phosphate group
  - Sugar
    - Ribose in RNA
    - Deoxyribose in DNA
  - Nitrogenous Base
    - C, T (only in DNA), U (Only in RNA)
    - A, G

Genetic Code

Defination

Sequence of nucleotides (nitrogen bases) in mRNA that contains information for protein synthesis (translation).

Features

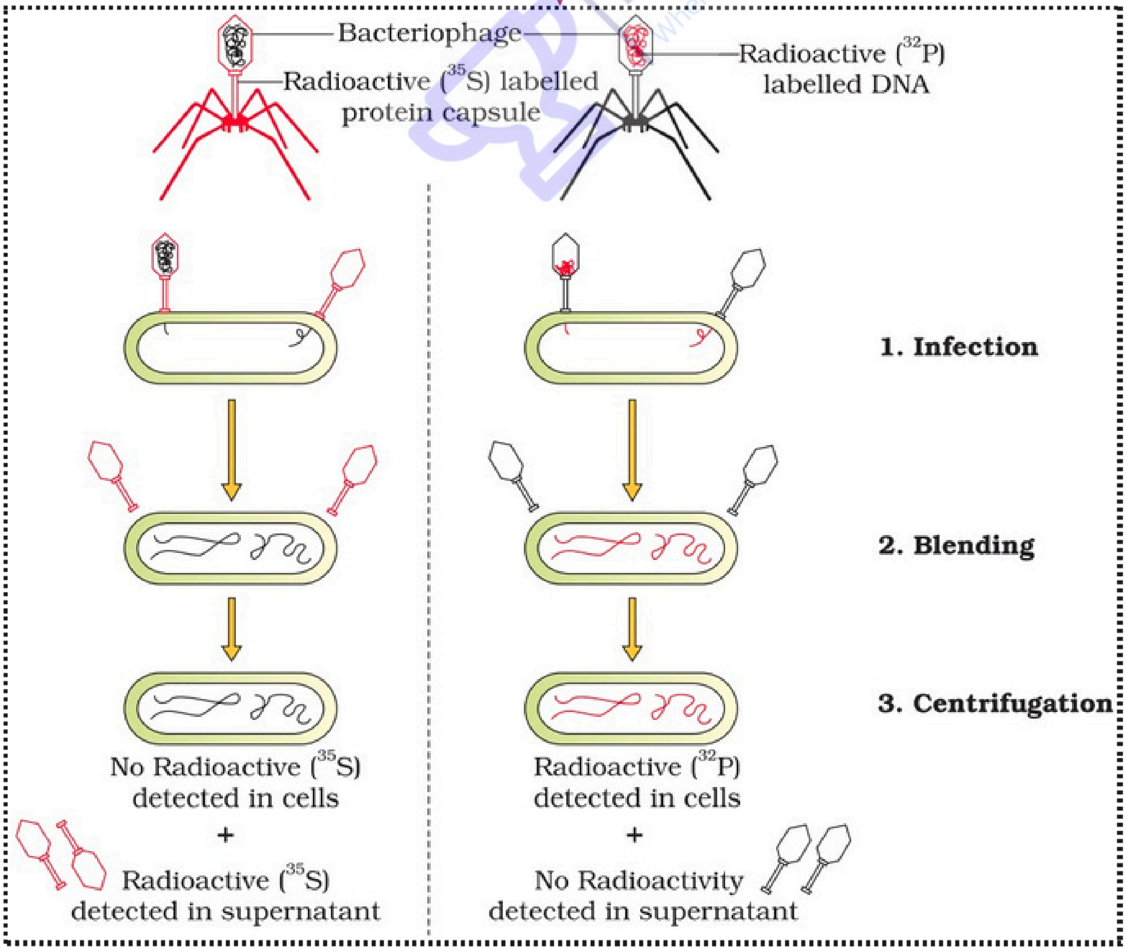
- Triplet
- Non- ambiguous
- Degenerate
- Comma less
- Universal
- Comma less

Human Genome Project

- To identify all the estimated genes in human DNA
- To determine the sequences of the 3 billion base pairs
- To provide tools for data analysis

Search for genetic material

Hershey and chase exp

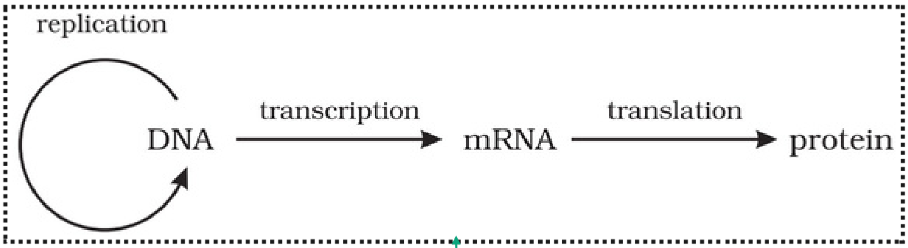


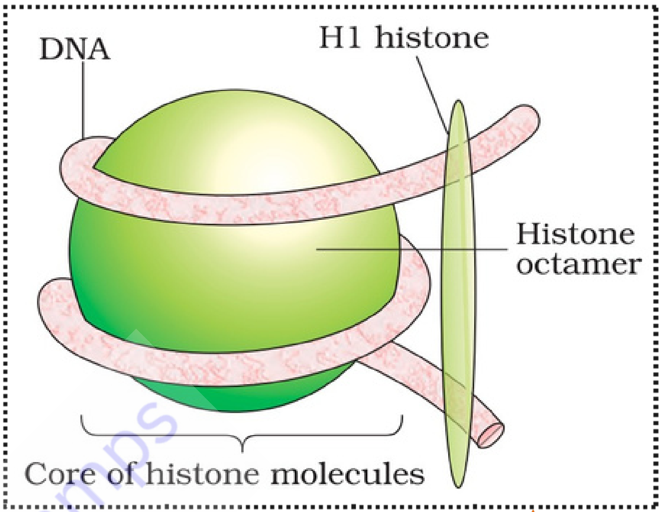
Biochemical Nature

- Discovered by Oswald Avery, Colin MacLeod and Maclyn McCarty.
- Conclusion: DNA is the hereditary material

Griffith Exp

- S- strain→ Inject into mice → Mice die
- R- strain→ Inject into mice → Mice live
- S- strain (Heat killed)→ Inject into mice → Mice live
- S- strain (Heat killed) + R- strain (live) → Inject into mice → Mice die
- Conclusion: Some 'transforming principle', transferred from heat- killed S- strain to R- strain.





Central Dogma

Gene expression and regulation

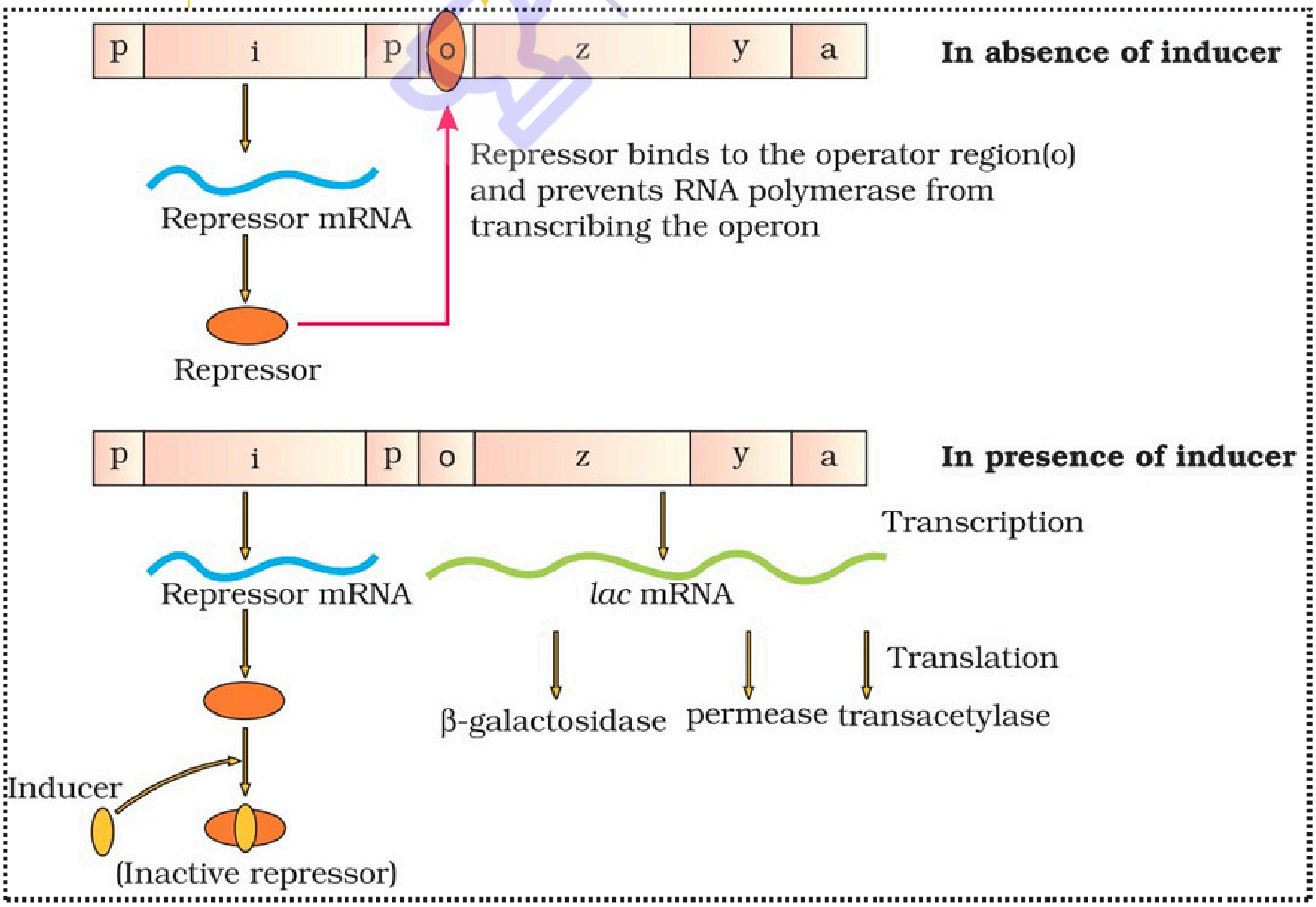
- Structural gene
- Operator gene
- Promoter gene
- Repressor
- Inducer

The operon controlling lactose metabolism

Lac Operon

Molecular Basis of Inheritance

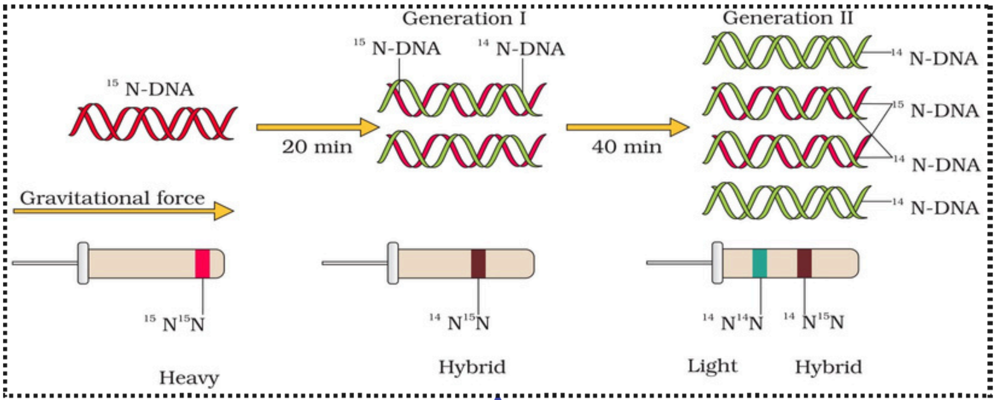
DNA packaging Nucleosome



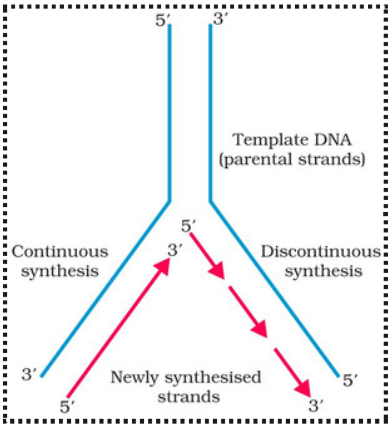
Molecular Basis of Inheritance

Replication

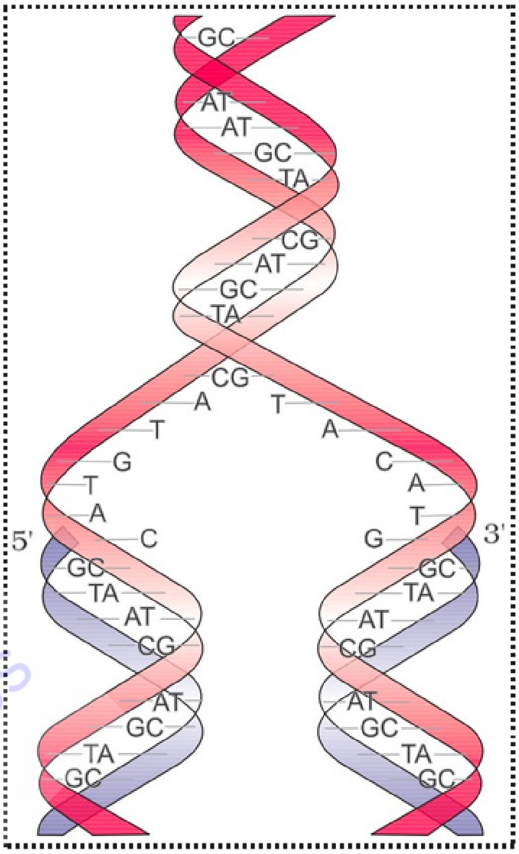
Meselson and stahl experiment



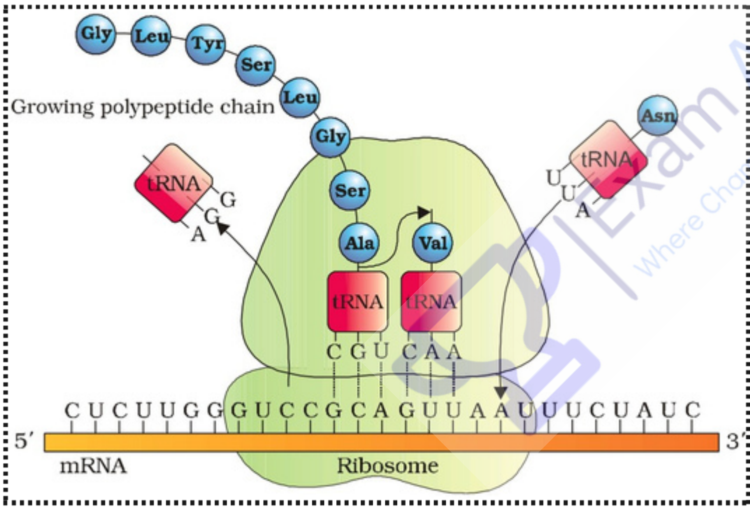
Process



Semi- conservative DNA Replication

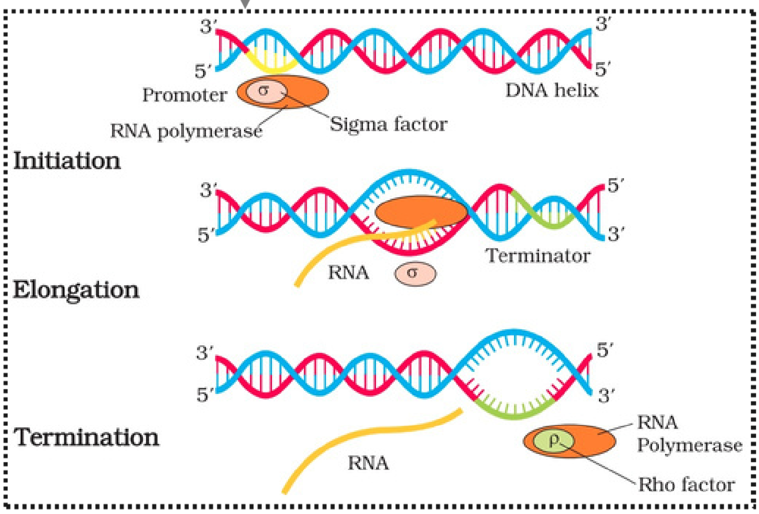


Translation

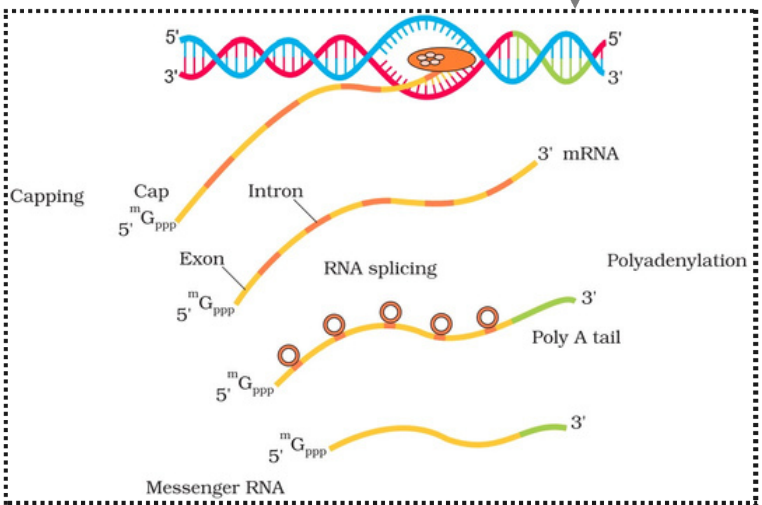


Transcription

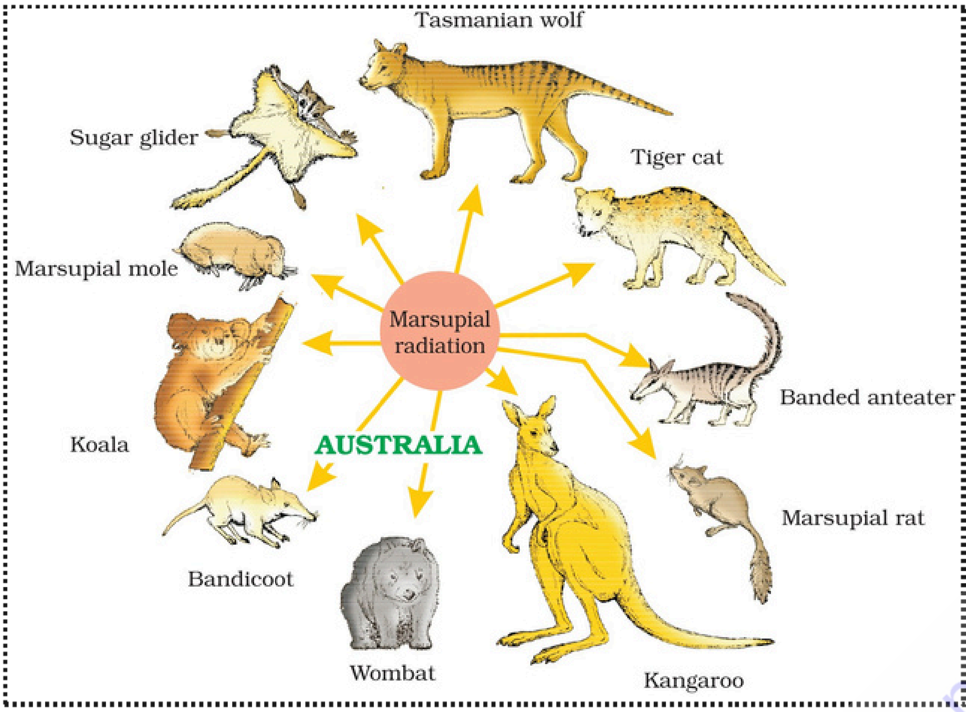
In Prokaryotes



In Eukaryotes



Class : 12th Biology
Chapter- 7 : Evolution



Evolution

Evidences

- **Bio Geographical, Adaptive Radiation**
  - Marsupial Radiation
  - Darwin Finches
  - Placental Mammals
    - E.g., Placental wolf and Tasmanian wolf- marsupial.
- **Morphological & Anatomical**
  - Homologous Organ
    - Human hand, Whale's flippers, Bat's wing, and Cheetah's foot.
    - Thorns of Bougainvillea and tendrils of Cucurbita.
  - Analogous Organ
    - Wings of insects and wings of birds. - Eyes of Octopus and mammals - Flipper of Penguins and Dolphins. - Sweet potato (modified root) and Potato (modified stem)
- **Palentological Evidence**
  - By study of fossils
  - Fossils help to study phylogeny, connecting link and about extinct animals
- **Bio- Chemical**
  - Similarities in proteins and genes.
- **Embryological**
  - Similarities in embryo development.

Theories of Biological Evidences

- **Lamarckism**
  - Proposed by Lamarck.
  - E.g. Long neck of giraffe.
- **Darwinism**
  - Proposed by Charles Darwin as Theory of Natural Selection

Human Evolution

Human ancestors	Origin period
Dryopithecus	20-25 mya
Ramapithecus	14-15 mya
Australopithecus	3-4 mya
Homo habilis	2 mya
Homo erectus (java man)	1.5 mya
Homo sapiens neander-thalesis (primitive man)	1,00,000-40,000 years ago
Homo sapiens sapiens (modern man)	75,000-10,000 years ago

Theories

- Theory of Panspermia
- Theory of Biogenesis
- Theory of Spontaneous Generation
- Theory of Chemical Evolution

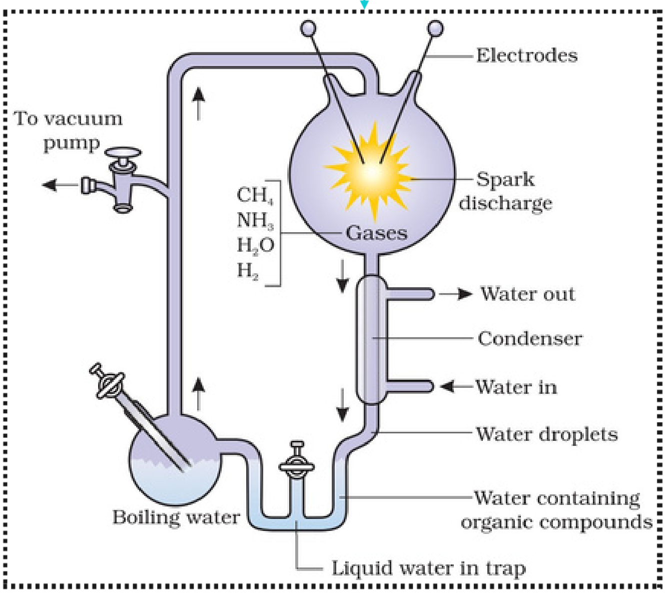
Hardy Weinberg

- Factors affecting
  - Gene Migration
  - Genetic drift
  - Mutation
  - Genetic recombination
  - Natural Selection

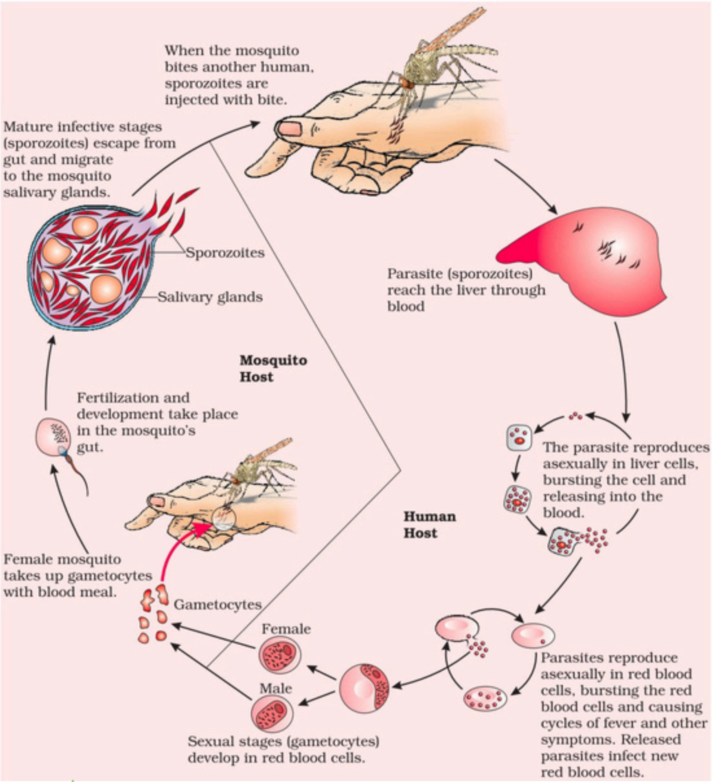
Mechanism

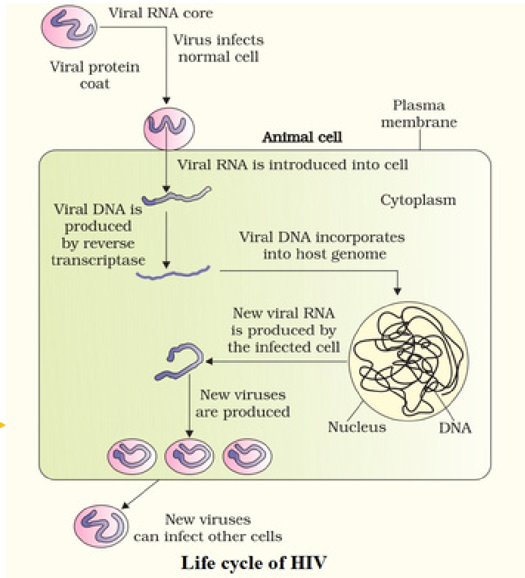
- Saltation by De Vries
  - Single Step Mutation
    - Proposed by Lamarck.
    - E.g. Long neck of giraffe.

Origin of Life- Urey and Miller's Experiment



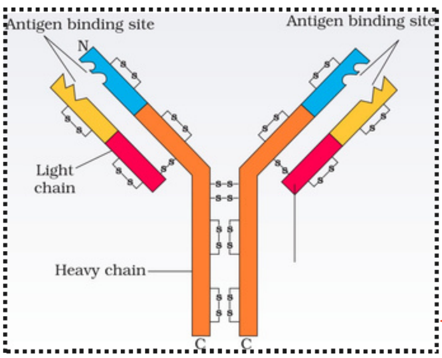
Class : 12th Biology
Chapter- 8 : Human Health and Diseases



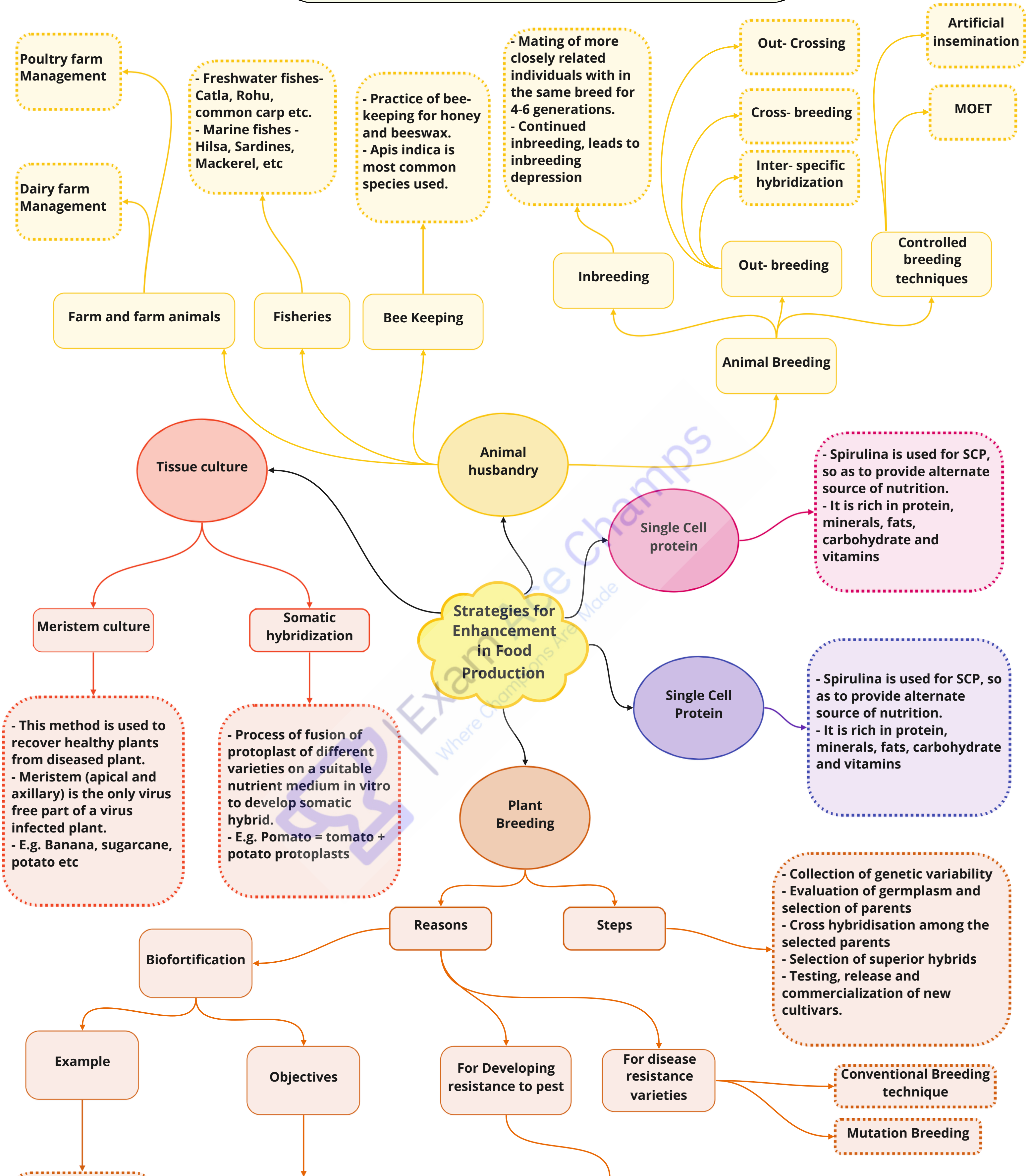


Human Health and Diseases

- **Health**
  - State of complete physical, mental and social well- being
- **Auto Immunity**
  - Memory based acquired immunity, which is able to distinguish foreign molecules or cells (pathogen) from self- cells.E.g. Rheumatoid arthritis, Multiple sclerosis
- **Allergy**
  - Exaggerated response of the immune system to allergens.
  - Antibodie s I g E type.
  - Anti-histamine, adrenaline and steroidsquickly reduce the symptoms.
- **AiDS**
  - Caused by HIV
- **Lymphoid organs**
  - Primary
    - Bone marrow, Thymus
  - Secondary
    - Spleen, Lymph node, Tonsils
- **Diseases**
  - Malaria
    - Plasmodium
  - Filariasis
    - Wuchereria
  - Ascariasis
    - Ascaris
  - Pneumonia
    - Streptococcus Pneumonia
  - Common Cold
    - Rhino virus
  - Typhoid
    - Salmonella typhi
  - Amoebiasis
    - Entoamoeba histolytica
- **Cancer**
  - Types
    - Benign tumor
    - Malignant
    - Metastasis
  - Detection & Diagnosis
    - Biopsy
    - Radiography
    - CT
    - MRI
  - Treatment
    - Radiation therapy
    - Chemotherapy
    - Immunotherapy
- **Immunity**
  - Pathogens
    - Types of barriers
      - Physical
      - Physiological
      - Cellular
      - Cytokine
  - Innate (Non- specific)
    - Produces primary & Secondary response
      - B Lymphocytes
        - Produces Antibodies
      - T Lymphocytes
        - Help B Cells
  - Acquired (Specific)
    - 2 Types
      - Active Own antibiotics
        - Natural: exposure to infectious agent
        - Artificial: immunization
      - Pasive Ready made antibiotics
        - Natural: maternal antibodies e.g., colostrum
        - Artificial: antibodies from other sources like anti- tetanus serum
- **Drugs and Acohol abuse**
  - Drugs
    - Opioids- Papavarsomniferum
    - Coca alkaloid- Erythroxylum coca.
    - Cannabinoids- Cannabis sativa
  - Symptoms and signs
    - Drop in academic performance
    - Lack of interest in personal hygiene.
    - Withdrawal and isolation.
    - Depression, aggressive and rebellious behavior.
    - Deteriorating relationship between family and friends.
    - Loss of interest in hobbies.
  - Causes
    - Curiosity and Experimentation. - Need for adventure and excitement.
    - Escape from facing problems.
    - Stress from pressure to excel in academics or examination.
    - Television, movies, newspapers, internet
    - Unsupportive family structures and Peer pressure
  - Prevention control
    - Avoid undue peer pressure.
    - Education and counselling.
    - Seeking help from parents and peers.
    - Looking for danger signs.
    - Seeking professional and medical help.



Class : 12th Biology
Chapter- 9 : Strategies for Enhancement in Food Production

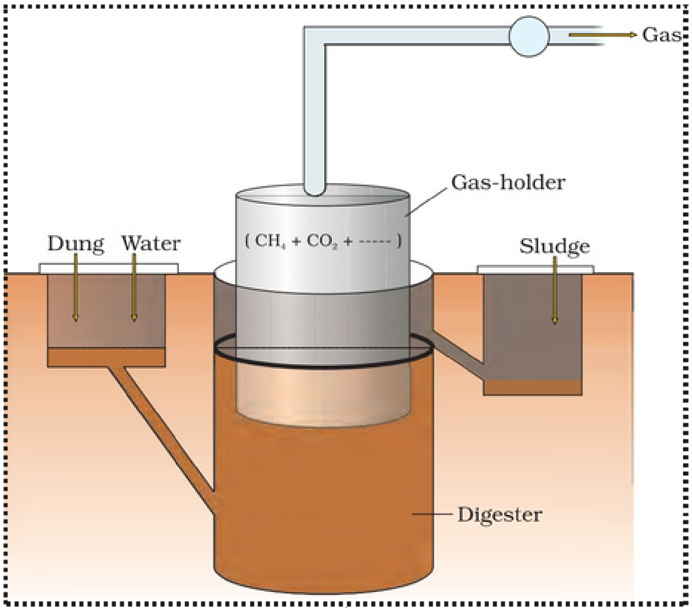


Crop	Variety	Insect Pests
Brassica (rapeseed mustard)	*Pusa Gaurav*	*Aphids*
Flat bean	*Pusa Sem 2, Pusa Sem 3*	*Jassids, aphids and fruit borer*
Okra (Bhindi)	*Pusa Sawani Pusa A-4*	*Shoot and Fruit borer*

Class : 12th Biology
Chapter- 10 : Microbes in Human Welfare

Microbes in Human Welfare

Microbes in Production of Biogas
- Bacteria involved is Methanobacterium



Microbes in household products
- Curd
  - Lactobacillus Or LAB
- Bread
  - Saccharomyces cerevisiae
- Toddy
- Swiss Cheese
  - Propionibacterium sharmanii

Microbes as biocontrol agents
- Bacillus thuringiensis: Control butterfly caterpillar
- Tichoderma sp.
- Baculoviruses (esp. genus Nucleopolyhedrovirus): Controls insect and arthropods

Microbes in Industrial Products
- Organic acids
  - Aspergillus niger Acetobacter aceti Clostridium butylicum Lactobacillus (a fungus): Citric acid (a bacterium): Acetic acid (a bacterium): Butyric acid (a bacterium): Lactic acid
- Fermented beverages
  - Saccharomyces cerevisiae
- Enzymes
  - Lipases
  - Pectinases and proteases (To clarify bottels juices)
  - Streptikinase (Streptococcus- used as clot buster)
- Antibiotic Penicillin
  - Penicillium notatum
- Bioactive molecules
  - Statins- Monascus purpureus used as blood-cholesterol lowering agents
  - Cyclosporine A- Trichoderma polysporum used as an immunosuppressive drugs

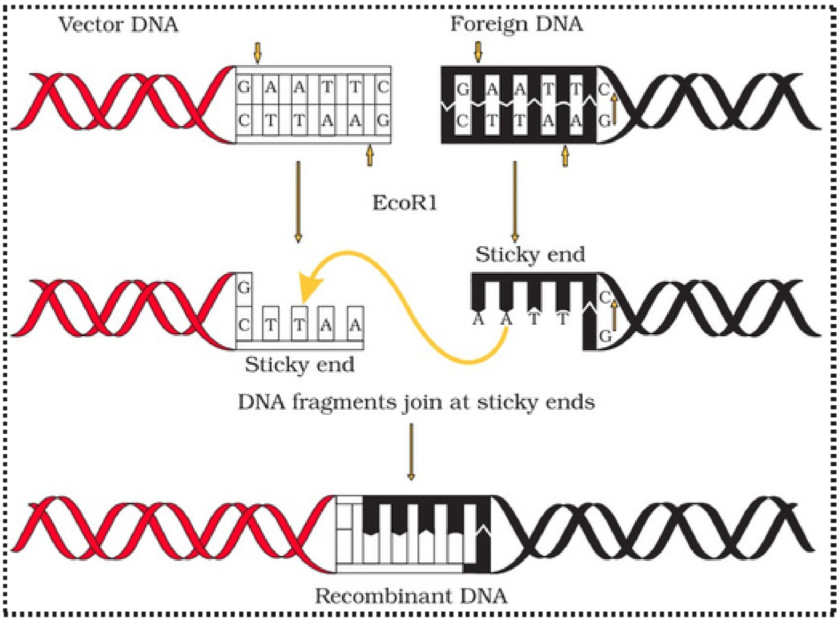
Microbes as bio- fertilizers
- Fungi
  - Mycorrhiza- Association of Glomus with plants
- Cynobacteria
  - Symbiosis: Anabaena in Azolla
  - Free living: Nostoc, Oscillatoria and blue green algae
- Bacteria
  - Rhizobium (Symbiotic bacteria)
  - Azospirillum and Azotobacter (free living bacteria)

Microbes in Sewage Treatment
- Secondary Sewage Treatment
  - Involves the action of microbes
- Primary Sewage Treatment
  - Involves removal of coarse solid materials

Biotechnology Principles and Processes

Tools of rDNA Technology

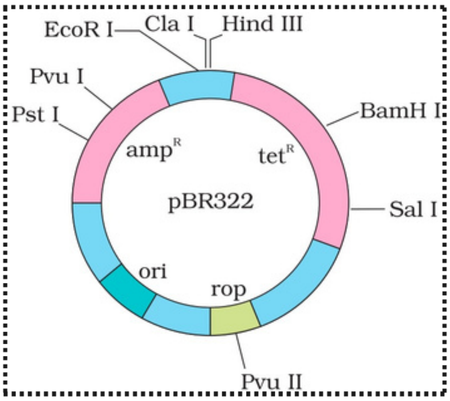
Restriction Enzymes



Cloning vector

Features of vector

- Has Ori
- Has Selectable Markers
- Has fewer cloning sites

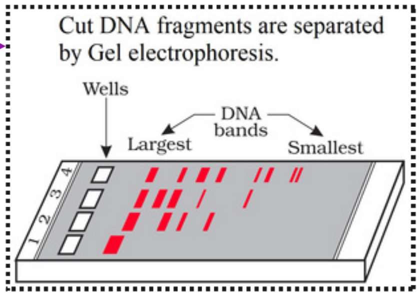


Competent host

Host is made competent by:

- **Biolistics** – Cell are bombarded with high velocity micro particles of gold or tungsten coated with DNA
- **Micro injection** – rDNA is directly injected into the nucleus of an animal cell
- **Disarmed pathogen vector** – Vectors like Agrobacterium when infects the cell transfer the recombinant DNA into the host

It is the integration of natural science and organisms, cells, parts thereof, and molecular analogues for product and services

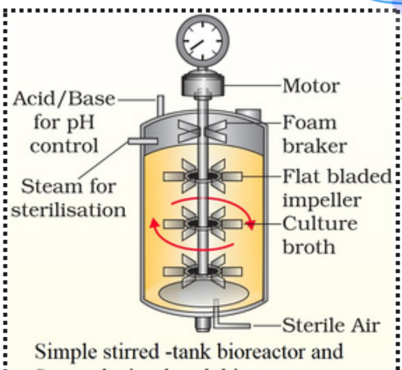


Principles of Genetic engineering

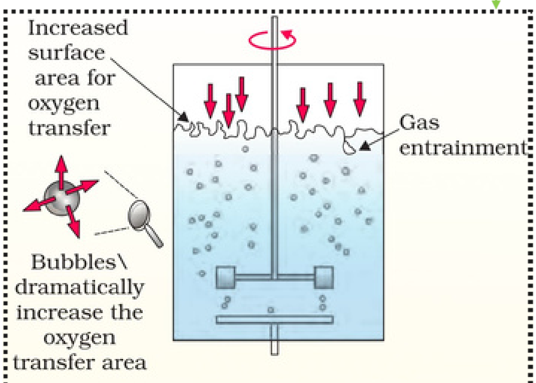
- rDNA
- Gene cloning
- Gene transfer

Processes of rDNA Technology

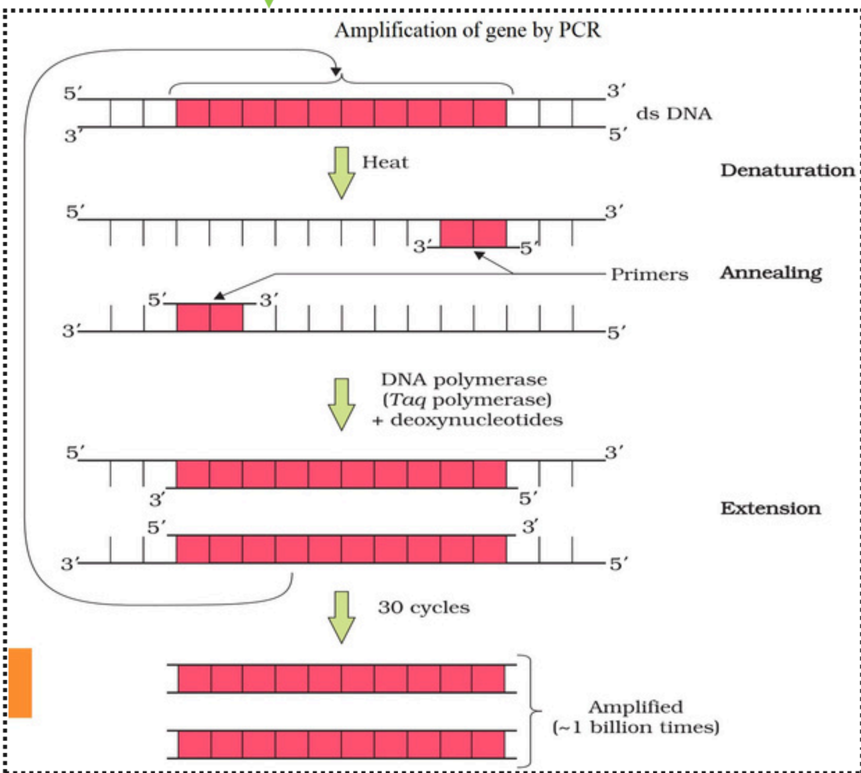
- Cutting of DNA at specific locations by restriction enzymes
- Transfer of rDNA into the Host cell
- Obtaining foreign gene product by bioreactors



Simple stirred -tank bioreactor and Sparged stirred-tank bioreactor



Obtaining foreign gene products by bioreactors



Class : 12th Biology
Chapter- 12 : Biotechnology and its Application

Biotechnology and its Application

In Agriculture

Advantages of GM Crops

- Makes crops more tolerant to abiotic stresses
- Reduces resilience on chemical pesticides.
- Helps to reduce post- harvest losses.
- Increases efficiency of mineral usage by plants.
- Enhances nutritional value of food. E.g. Golden rice

Pest resistance in cotton plant

- Bacillus thuringiensis contain Bt toxin, which is in inactive form. - It becomes active toxin in the alkaline pH of insect gut. - The activated toxin binds to the surface of midgut epithelial cells of insect and creates pores. - This causes cell swelling and lysis and eventually death of the insect.

Pest resistance in Tobacco plant

RNA interference (RNAi) strategy is used to prevent the infestation of tobacco plants by Nematode (Meloidegyne incognitia).

GEAC

- Make decisions about the validity of GM research.
- Safety of GM- organisms for public services.

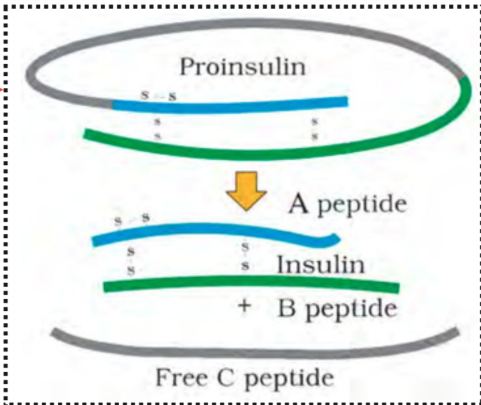
Biopiracy and biopatent

- Use of bio- resources by multinational companies and other organizations without proper authorization from the countries and people concerned is biopiracy.
- Patents granted for biological entities and for products derived from them are called biopatents.E.g. Basmati Rice.

Gene Therapy

- Technique for correcting a defective gene through gene manipulation.
- First clinical gene therapy was given in 1990 to a 4- year old girl with adenosine deaminase (ADA) deficiency.

In Medicine Genetically Engineered insulin



Molecular diagnosis

- Polymerase Chain Reaction.
- Enzyme Linked Immuno- sorbent Assay (ELISA)

Transgenic animals

Definition

- Animals that have their DNA manipulated to possess or express an extra gene.
- E.g. Transgenic rats, rabbits, pigs, sheep, Rosie cow and fish.

Advantages

- To study normal physiology and development.
- To study the contribution of genes in the
development of a disease.
- Biological products.
- Vaccine safety testing.
- Chemical safety testing

Organisms and Populations

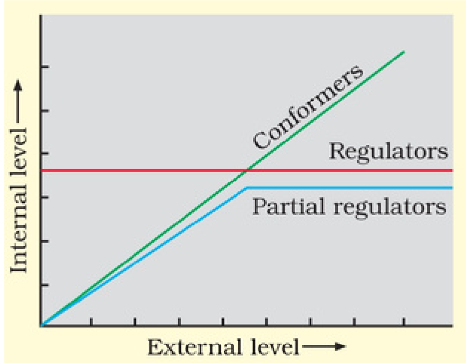
Organisms and its Environment

Adaptation

Attribute of organism's morphological, physiological and behavioural changes that enables the organisms to survive and reproduce in its habitat

- Adaptation in High Altitude in humans
  - People suffer from mountain sickness
  - Relief occur due to acclimatization: Increasing breathing rate, increasing RBCs production
- Adaptation in Lizards
  - Absorb heat
  - Move into shade
- Adaptation in Mammals
  - Allen's rule
  - Thick layer of fat (as in seal)
- Adaptation in Desert plant
  - Thick cuticle on leaf
  - CAM
  - No leaves
- Adaptation in kangaroo rat
  - Meeting water requirement by fat oxidation
  - Produce concentrate urine

Organismic responces



Response to abiotic factors
- Regulate
- Conform
- Migrate - Suspend

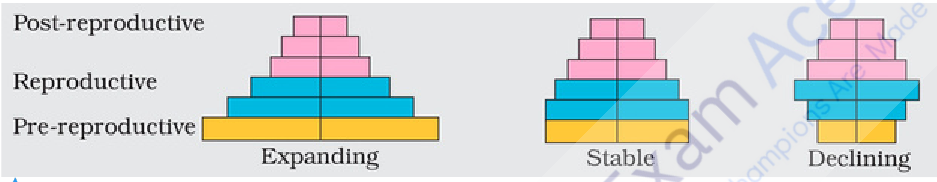
Major abiotic factors
- Temperature
- Water
- Light
- Soil

Population Interaction

Species A	Species B	Name of Interaction
+	+	Mutualism
–	–	Competition
+	–	Predation
+	–	Parasitism
+	0	Commensalism
–	0	Amensalism

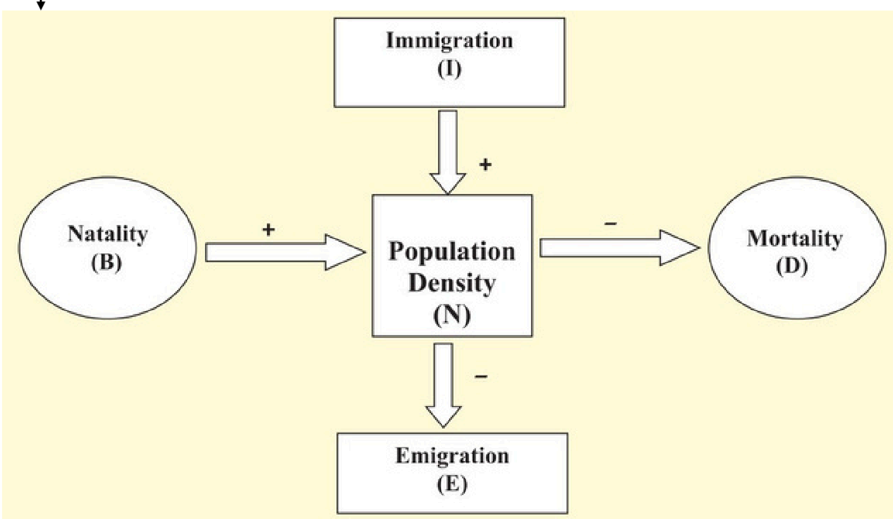
Population

Population Atributes
- Age Pyramids

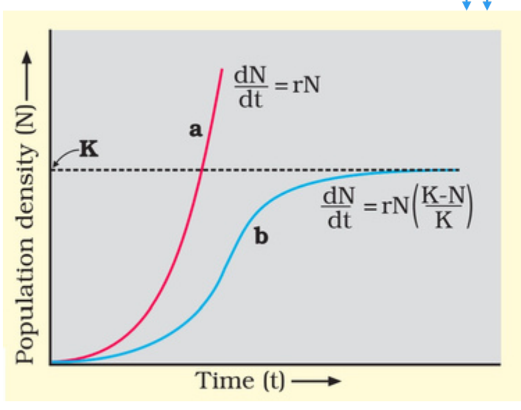


- Natality rate
- Mortality rate
- Sex Ratio
- Population Size

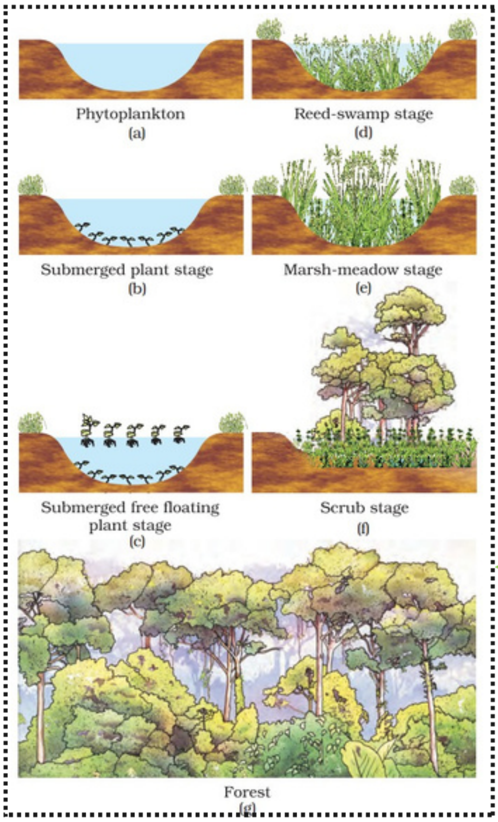
Population Growth
- Population Density
  - Natality
  - Mortality
  - Immigration
  - Emigration
- Growth Models
  - Population Density eqn.: $N_{t+1} = N_t + [(B+I)-(D+E)]$



  - Logistic Growth: $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$
  - Exponential Growth: $N_t = N_0 e^{rt}$



Ecosystem



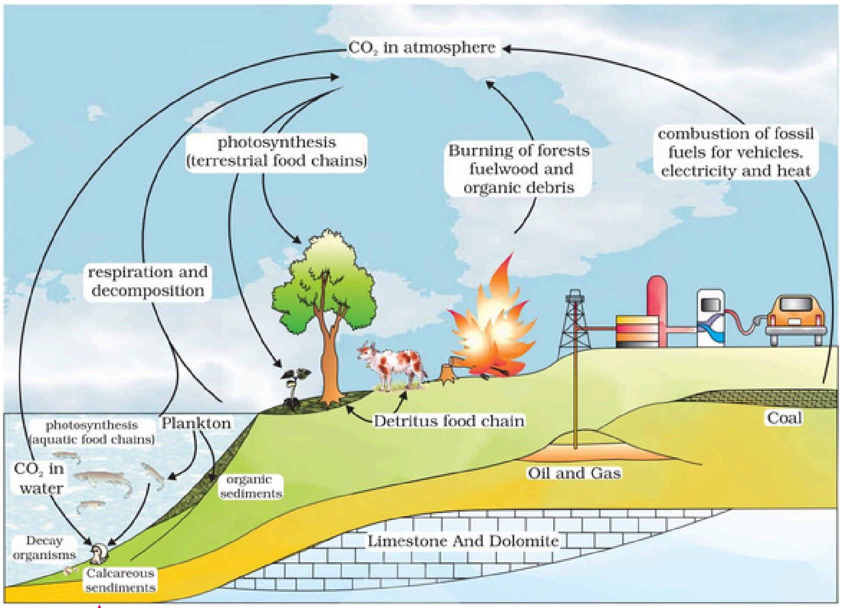
Primary Succession

Ecological Services
- Purify air and water.
- Mitigate droughts and flood.
- Cycle nutrients.
- Generates fertile soil.
- Provide wildlife habitat.
- Maintain biodiversity.
- Pollinate crops.
- Provide storage site for carbon
- Provide aesthetic, cultural and spiritual values.

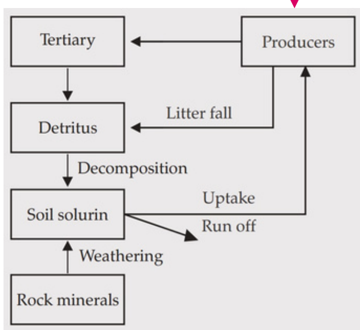
Microbes as biocontrol agents

Nutrient Cycling

Carbon Cycling

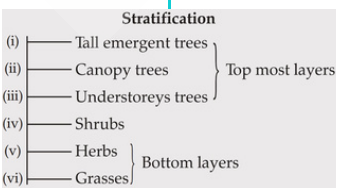


Phosphorus Cycling



Structure and function Ecosystem

- **Comprises**
  - **Biotic**: Autotrophs, Herbivores and carnivores
  - **Abiotic**: Pond, lake, river, estuary
- **Distribution of different species at different levels**



Functional Components of Ecosystem

- **Energy Flow**
  - Food chain shows energy pathways. Energy passes from producer (plants) to Plant- eaters, to Animal- eaters and to tertiary consumer
    - **Ten Percent Law**: During transfer of energy from one trophic level to another, only about 10% is stored at higher levels; remaining 90% is lost in respiration (heat).
    - **Trophic level**: Specific place occupied by an organism in the food chain in an ecosystem
    - **Food Chain**
      - **Detritus food chain**: Begins with detritus such as dead bodies of animals or fallen leaves.
      - **Grazing food chain**: Begins with producers, present at the first trophic level.
    - **Ecological Pyramid**
      - **Pyramid of Energy**: Always upright because energy is lost as heat at each step
      - **Pyramid of Biomass**: - Upright e.g. in grassland - Inverted e.g. in pond
      - **Pyramid of No.**: - Upright e.g. in grassland - Inverted e.g. in tree ecosystem
- **Process of Decomposition**
  - Fragmentation
  - Leaching
  - Catabolism
  - Humification
  - Mineralisation
- **Productivity**
  - **Primary**: NPP= GPP- R
  - **Secondary**: Rate of formation of new organic matter by consumers
- **Nutrient cycling**

Class : 12th Biology
Chapter- 15 : Biodiversity and Conservation

Biodiversity and Conservation

Conservation of Biodiversity

Reasons

- **Narrow Utilitarian**
  - More than 25% of the drugs are derived from plants.
- **Broadly Utilitarian**
  - Provides oxygen
  - Pollination
  - Aesthetic pleasures
- **Ethical Utilitarian**
  - Moral duty to take care for biodiversity

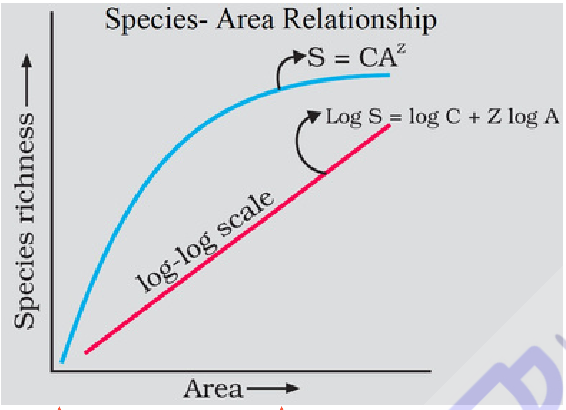
Methods

- **In- situ Conservation**
  - National Park
  - Sanctuary
  - Biosphere Reserves
  - Sacred Forests
  - Hotspots
- **Ex- situ Conservation**
  - Zoological Parks
  - Botanical gardens
  - Gene banks
  - Genetic resource centres

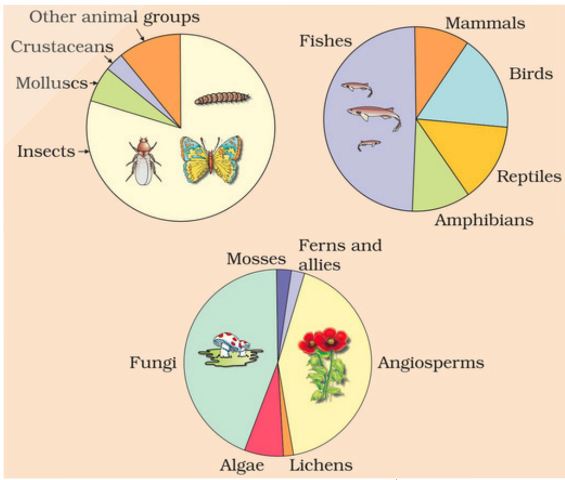
Biodiversity

Species- Area Relationship

log S = log C + Z log A



Global Species Diversity



Levels of biodiversity

- **Genetic diversity**
  - Diversity show n by a single species at genetic level.
- **Species diversity**
  - Diversity at the species level
- **Ecological diversity**
  - Diversity at the ecosystem level

Patterns of biodiversity

- **Latitudinal gradients**
  - Species diversity decreases from the equator towards the poles.
  - Tropics (latitudinal range of 23.5° S to 23.5° N) have more species than temperate o or polar areas.

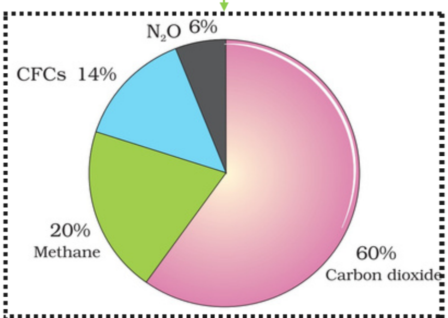
Causes of loss of biodiversity

- Habitat loss and fragmentation
- Over- exploitation
- Alien- species invasions
- Co- extinctions

Class : 12th Biology
Chapter- 16 : Environmental Issues

Environmental Issues

- **Air Pollution**
  - Smoke stacks from thermal plants, garbage decomposition, burning of fossil fuels are common sources.
  - Can be controlled by electrostatic precipitator, scrubber and catalytic converters
- **Global Warming**
  - **Causes**: increased concentration of greenhouse gases
  - **Effects**:
    - Odd climatic changes (El nino effect).
    - Melting of Polar ice caps.
    - Rise in sea level.
  - **Methods of control**:
    - Reducing the use of fossil fuels
    - Use of bio- fuels
    - Improving energy efficiency
    - Use of renewable source of energy such as CNG etc.
- **Water Pollution**
  - **Effects**
    - Biomagnification
    - Eutrophication
    - Increased BOD
  - **Sources**
  - **Intergrated waste water Management**
    - Treat Wastewater in integrated manner. E.g., in town of Areata situated along Northern coast of California.
    - Use of Ecosan toilets for ecological sanitation
- **Deforestation**: Conversion of forested areas to non- forested areas
- **Noice Pollution**: Noise is undesirable high level of sound.
- **Solid Wastes**
  - Types: Biodegradable and non- biodegradable.
  - Disposal methods are sanitary landfills, rag pickers, Incineration, recycling
  - **Remedy of plastic waste**: Ahmed Khan developed Polyblend. It is a fine powder of recycled modified plastic used to lay roads.
- **Zone Depletion**
  - CFCs (used as refrigerants) causes ozone depletion. It has formed ozone hole over the Antarctic region
  - Montreal Protocol was signed to control the emission of ozone depleting substances
- **Green House effect**
  - **Green house effect**



  - **Definition**: Naturally occurring phenomenon, responsible for heating of earth's surface and atmosphere.
- **Redioactive waste**
  - Wastes generated from radioactive materials.
  - Nuclear wastes should be buried in rocks.